Erwann CARN

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Research interests: Robotics, Reinforcement Learning, Motion Retargeting and Imitation Learning.

Education

École Nationale des Ponts et Chaussées (IP Paris) — Champs-sur-Marne, France
MSc in Applied Mathematics & Computer Science (Learning & Vision track) — Sep. 2024 – Jul. 2027

- Relevant coursework: Optimal Control, Computer Vision, Deep Learning, Machine Learning, Probability, Stochastic Calculus, Linear Algebra, Statistics

Lycée Clemenceau — Nantes, France
*Preparatory Classes – MPSI/PSI** — Sep. 2022 – Aug. 2024

- Main coursework: Mathematics, Physics, Engineering & Computer Science

Publications

OTRetarget: Joint Robot and Object Motion Retargeting via Optimal Transport
G. Besset*, **E. Carn***, T. Carecchio, V. Tordjman–Levavasseur, F. Schramm, Y. de Mont-Marin, J. Carpentier, A. S. Sathya
Preprint – arXiv:2609.36602, 2026 — *Equal contribution

Research Experience

INRIA Paris (Willow team) — Paris, France
Robotics & AI Research Intern – Supervised by J. Carpentier — Jun. 2026 – Dec. 2026

- Co-developed OTRetarget, a contact-aware method jointly retargeting humanoid and object motion
- Retargeted 6,400+ motions (crawling, spins, object carrying) and transferred demos to unseen objects via optimal transport
- Trained RL tracking policies on the references and deployed them on a Unitree G1 and a Shadow Hand E
- Currently working on faster RL training for G1 locomotion with high update-to-data off-policy methods (FlashSAC)

CEREA / École des Ponts — Paris, France
Research Project – ML for Weather Prediction – Supervised by S. Cheng — Jan. 2026 – Jun. 2026

- Developed ML models to reconstruct high-resolution weather fields from sparse and mobile observation networks
- Combined ERA5 reanalysis data with irregular satellite trajectories and ground-based sensors, comparing deterministic and generative approaches

Research & Academic Projects

Nextrash – Smart Waste-Sorting Bin — Jan. 2025 – Sep. 2026
Co-founder

- Designed a prototype with a CNN (TensorFlow) for real-time waste classification; secured €3,000 in funding

Computer Vision Projects — Sep. 2025 – Dec. 2025
Course-based, ENPC

- Camera pose estimation from homographies for augmented reality, with 3D tracking across views
- Harris corner detection from scratch and robust keypoint matching between images
- Patch-based texture synthesis and object removal by inpainting

Tennis Ball Impact Detection — Sep. 2022 – Aug. 2024
Research project

- Low-cost Hawk-Eye alternative using signal processing and trilateration (∼6 cm localization error)

Skills & Interests

Technical Skills: Python (PyTorch, NumPy, SciPy, scikit-learn, TensorFlow), C++, Git, LaTeX; Robotics (Pinocchio); RL (PPO, SAC, MuJoCo, IsaacSim, IsaacLab)

Languages: French (Native), English (Fluent), Spanish (Intermediate)

Extra-curricular: Junior Entreprise – Commercial Manager; IA student association – Treasurer

Interests: Basketball (Regional League), Running, Chess (1200+ Elo), DJing